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A Privacy-Preserving and Interpretable Multimodal Federated Learning Framework for Alzheimer's Disease Diagnosis

FIRST REVIEW

Guided by,

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Presented by,

SREELAKSHMI V S

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Vision of the Department

Creating eminent and ethical leaders in the domain of Computational Sciences through quality professional education with a focus on holistic learning and excellence.

Mission of the Department

- To create technically competent and ethically conscious graduates in the field of Computer Science and Engineering by encouraging holistic learning and excellence.
- To prepare students for careers in Industry, Academia and the Government.
- To instill Entrepreneurial Orientation and research motivation among the students of the department.
- To emerge as a leader in education in the region by encouraging teaching, learning, industry and societal connect.

Programme Educational Objectives (PEOs)

1. The graduates shall have sound knowledge of Mathematics, Science, Engineering and Management to be able to offer practical software and hardware solutions for the problems of industry and society at large.

2. The graduates shall be able to establish themselves as practicing professionals, researchers or Entrepreneurs in computer science or allied areas and shall also be able to pursue higher education in reputed institutes.

3. The graduates shall be able to communicate effectively and work in multidisciplinary teams with team spirit demonstrating value driven and ethical leadership.

Programme Outcomes

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSOs)

On the completion of Computer Science & Engineering program, the students will possess:

1. An ability to apply knowledge of data structures and algorithms appropriate to computational problems.
2. An ability to apply knowledge of operating systems, programming languages, data management, or networking principles to computational assignments.
3. An ability to apply design, development, maintenance or evaluation of software engineering principles in the construction of computer and software systems of varying complexity and quality.
4. An ability to understand concepts involved in modeling and design of computer science applications in a way that demonstrates comprehension of the fundamentals and trade-offs involved in design choices.

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- Architecture Diagram
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- References

Introduction

- Alzheimer's Disease (AD) is a progressive neurodegenerative disorder affecting cognitive functions.
- Early and accurate diagnosis is essential for effective clinical intervention.
- Artificial Intelligence (AI) techniques support automated and reliable AD prediction.
- Medical data privacy and model interpretability remain major challenges

Introduction

Phase 1

- Multimodal dataset collected from OASIS-3
- Integration of clinical, psychological, and MRI segmentation features
- Random Forest Classifier selected for prediction
- Achieved reliable accuracy and robustness
- Federated Learning adopted for privacy-preserving model training
- SHAP used for explainable AI and feature importance analysis

Introduction

Phase 1

- Extension of Phase-1 using raw MRI image datasets
- CNN-based deep learning for spatial and structural brain pattern learning
- Enhanced multimodal data utilization with improved diagnostic accuracy
- Grad-CAM integration for explainable and transparent predictions
- Privacy-preserving, interpretable, and scalable framework for real-world use

Problem Statement

- Alzheimer's diagnosis requires analysis of multimodal medical data
- Centralized machine learning approaches compromise patient data privacy
- Many existing AI models act as black boxes and lack explainability
- Feature-based methods may lose important spatial information from MRI images
- There is a need for a privacy-preserving, accurate, and explainable diagnostic framework

Objective

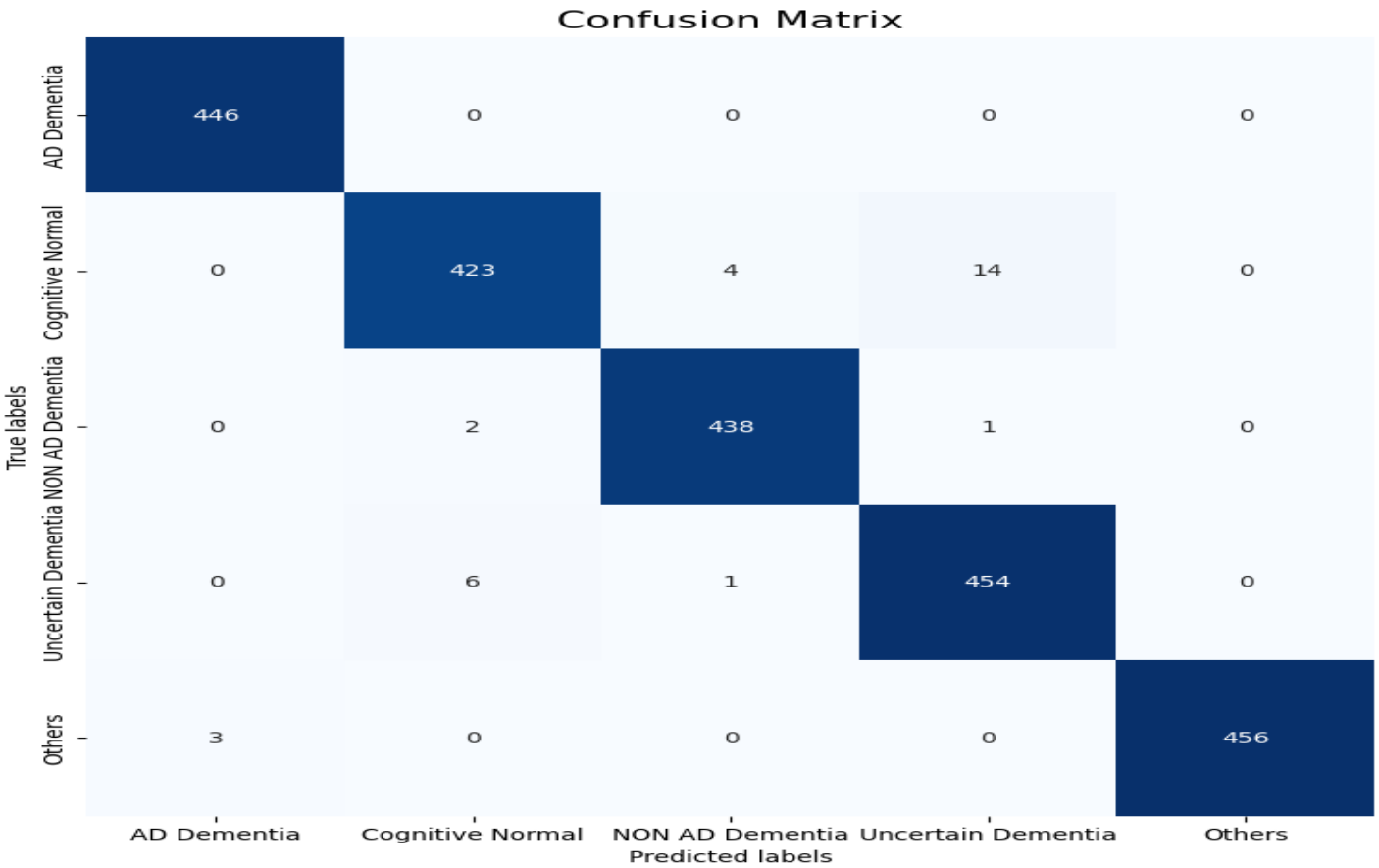
Phase 1	Phase 2
❖ To develop a multimodal Alzheimer's Disease prediction model using clinical, psychological, and MRI-derived features	❖ To enhance AD prediction accuracy by incorporating raw MRI image datasets
❖ To evaluate the effectiveness of traditional machine learning classifiers for AD prediction	❖ To design and implement CNN-based deep learning models for automatic feature learning

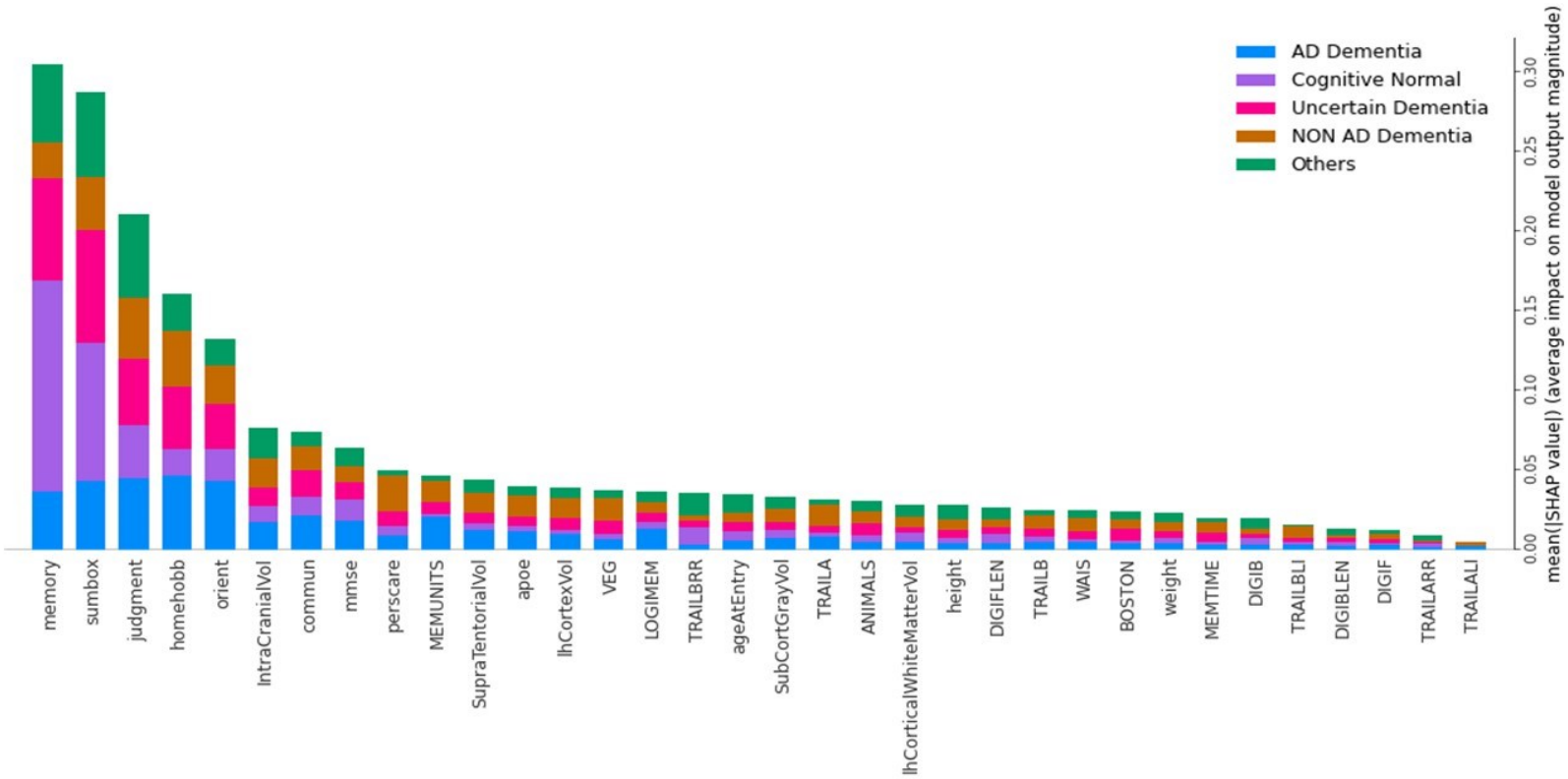
Phase 1	Phase 2
❖ To implement a Random Forest–based model with reliable performance	❖ To capture spatial and structural brain patterns associated with Alzheimer's Disease
❖ To ensure data privacy using a Federated Learning framework	❖ To develop a scalable, privacy-preserving, and clinically reliable AI framework for real-world application
❖ To improve model transparency using SHAP-based explainable AI techniques	❖ To improve interpretability using Grad-CAM visual explanation

Result Phase 1

- Achieved reliable Alzheimer's disease classification
- Preserved data privacy by avoiding centralized data storage
- Multimodal data improved prediction performance
- SHAP provided clear feature-level explanations
- Validated the feasibility of federated explainable learning

Result Phase 1




AD Dementia
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NON AD Dementia
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mean(|SHAP value|) (average impact on model output magnitude)
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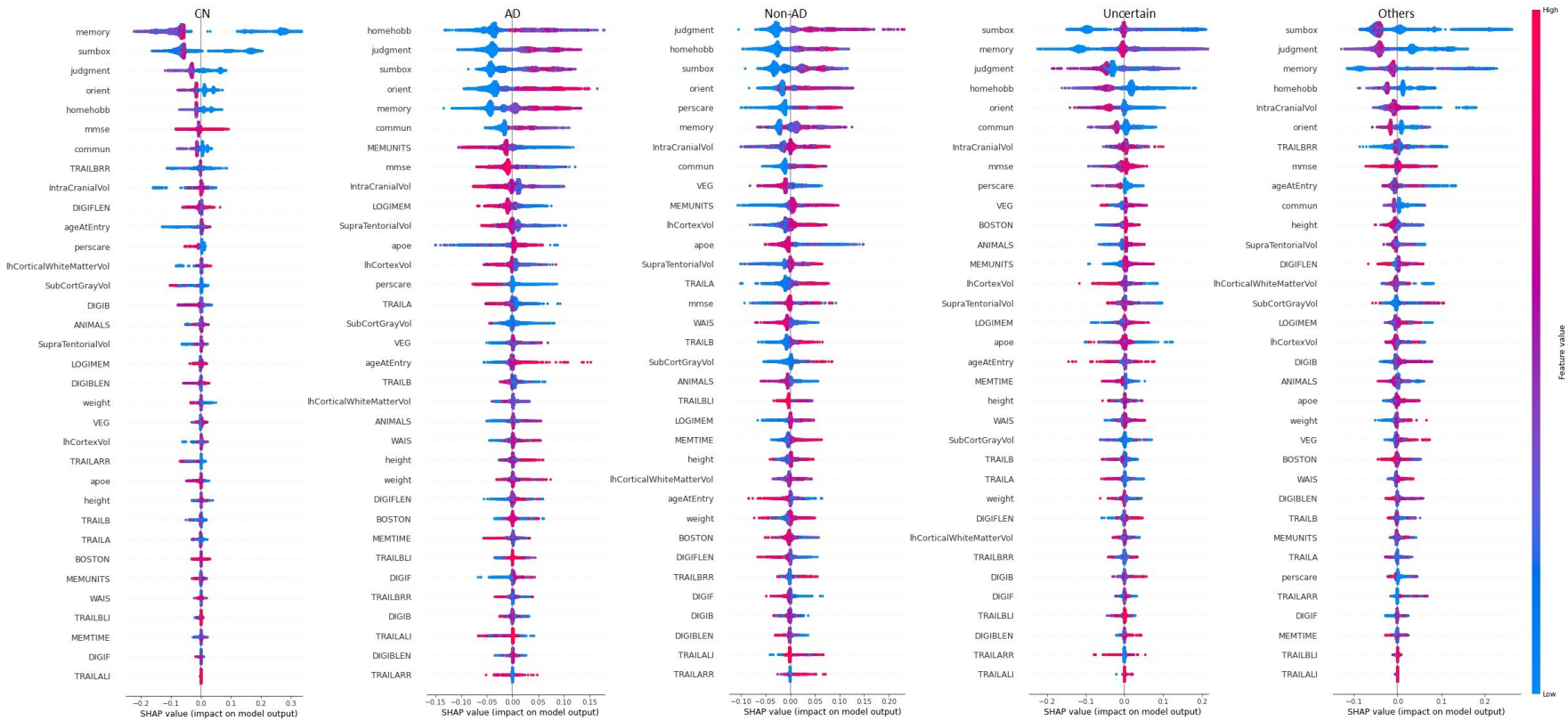
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SHAP value (impact on model output)
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IntraCranialVol
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SupraTentorialVol
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lhCorticalWhiteMatterVol
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SHAP value (impact on model output)
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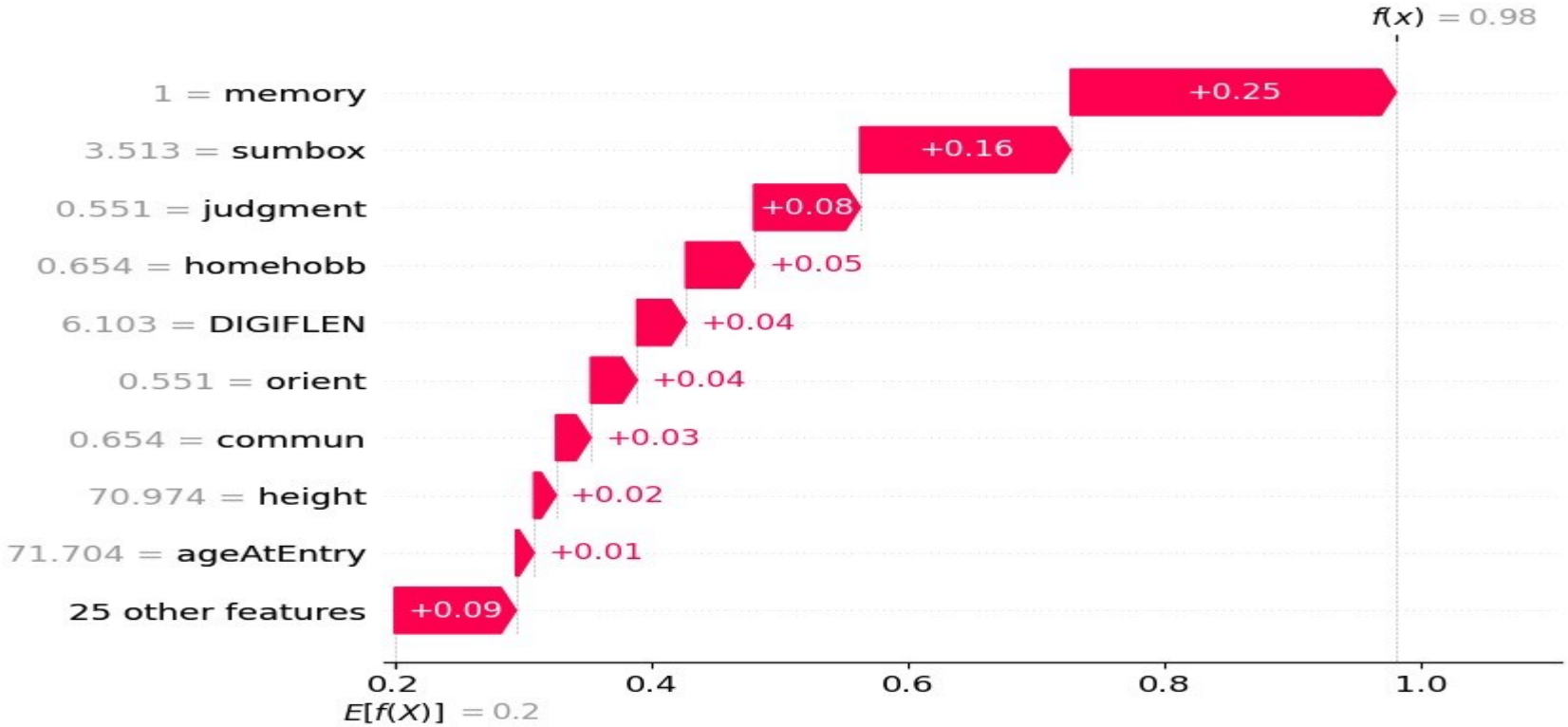
Feature value
Low


f(x) = 0.98
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3.513 = sumbox
0.551 = judgment
0.654 = homehobb
6.103 = DIGIFLEN
0.551 = orient
0.654 = commun
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71.704 = ageAtEntry
25 other features
+0.25
+0.16
+0.08
+0.05
+0.04
+0.04
+0.03
+0.02
+0.01
+0.09
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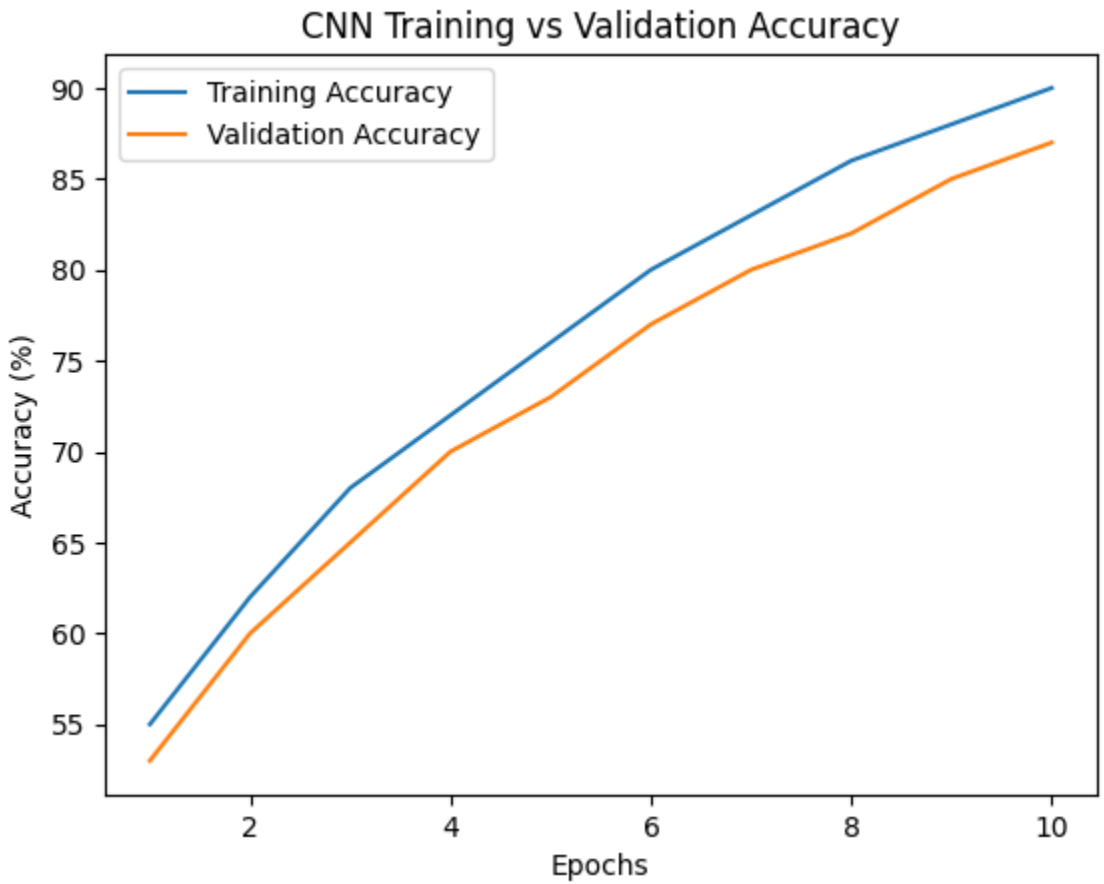
0.6
0.8
1.0
E[f(X)] = 0.2


CNN Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy (%)
90
85
80
75
70
65
60
55
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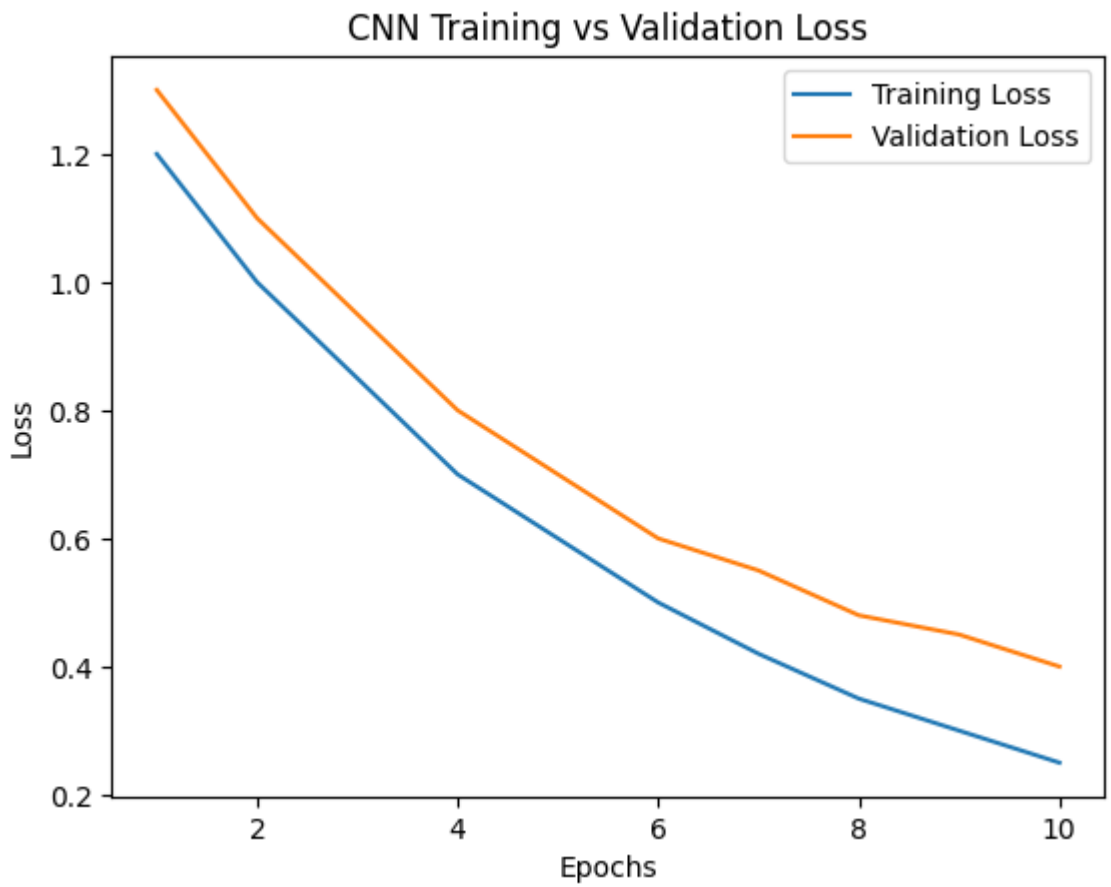
4
6
8
10
Epochs


CNN Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs
1.2
1.0
0.8
0.6
0.4
0.2
2
4
6
8
10

Work to be done

- Collection and preprocessing of raw MRI image datasets
- Development of CNN-based deep learning model for AD prediction
- Integration of multimodal data within a Federated Learning framework
- Implementation of Grad-CAM for visual explainability
- Performance evaluation and comparison with Phase-1 results

Work to be done

Feature ML → CNN / Deep Learning

- Manual feature extraction replaced by automatic feature learning
- CNN learns complex patterns directly from images

Classical FL → Deep Federated Learning

- Shallow models upgraded to federated CNN training
- Ensures privacy with high diagnostic accuracy

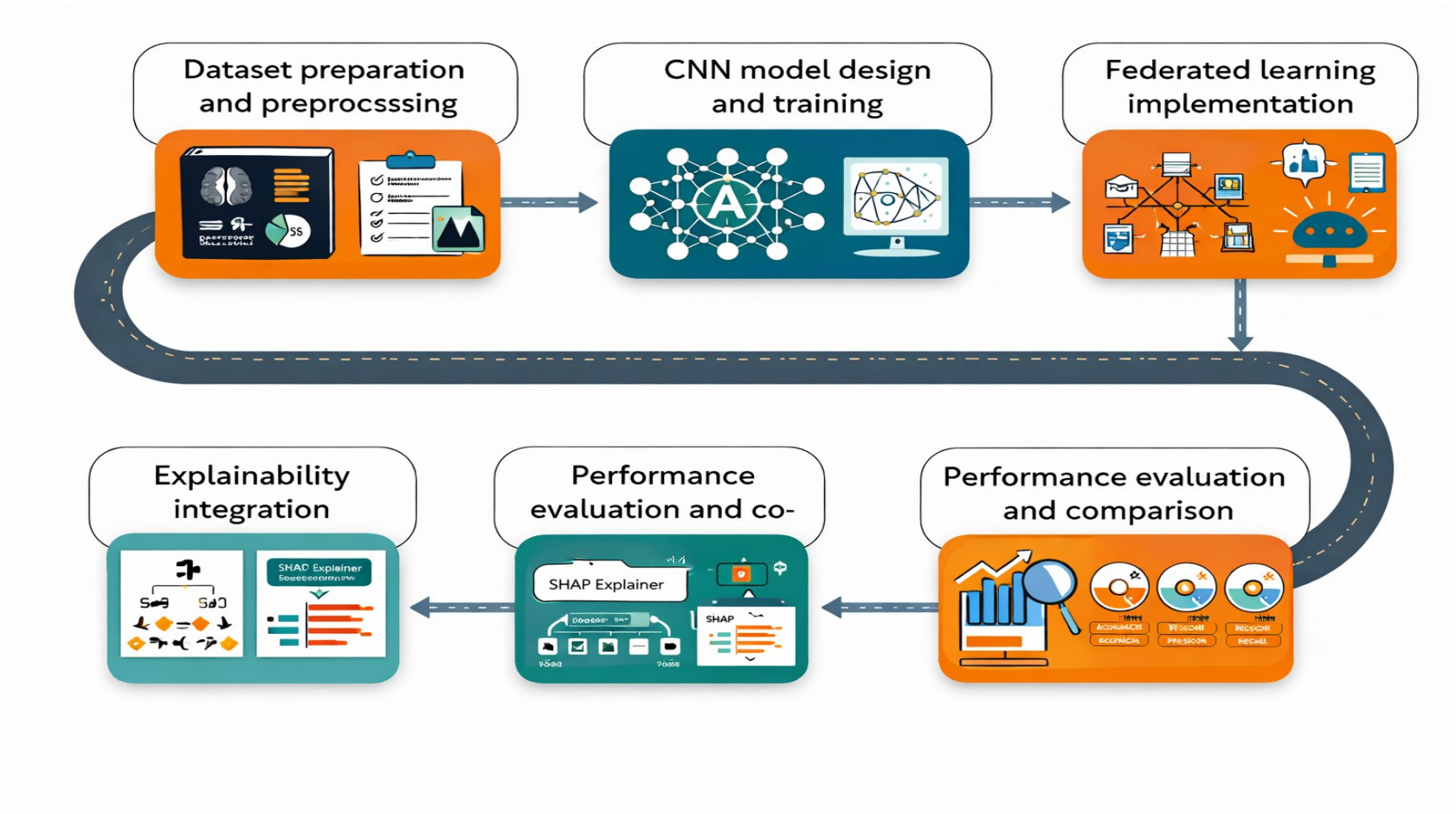
SHAP → Grad-CAM

- Feature-based explanation replaced by image-level heatmaps
- Highlights important regions in medical images

Road Map

- Dataset preparation and preprocessing
- CNN model design and training
- Federated learning implementation
- Explainability integration
- Performance evaluation and comparison

Road Map


Dataset preparation and preprocsssing
CNN model design and training
Federated learning implementation
Performance evaluation and comparison
Performance evaluation and co-
SHAP Explainer
SHAP
Explainability integration
SHAP Explainer

Literature Survey

Paper Name	Author	Year	Description	Technique
Federated Explainable AI-Based Alzheimer's Disease Prediction With Multimodal Data	S. Jahan et al.	2025	Proposes a privacy-preserving, interpretable AD prediction system using multimodal data	Federated Learning + Random Forest + SHAP (XAI)
Multimodal Alzheimer's Disease Prediction Using Data-Level Fusion	S. Jahan et al.	2023	Uses multimodal fusion of MRI, clinical & psychological data	Random Forest + SMOTE + SHAP
"Multiple Machine Learning Models for Detection of Alzheimer's Disease Using OASIS Dataset"	Baglat et al.	2020	Early AD detection using MRI features	ML models

Literature Survey

Paper Name	Author	Year	Description	Technique
"Comparison Between Explainable AI Algorithms for Alzheimer's Disease Prediction Using EfficientNet Models"	Jahan et al. [9]	2023	Comparison of explainable CNNs	EfficientNet
"BrainCrossFed CNN Model for Alzheimer Classification Using MRI Data and Comparison and Benchmarking with DINOv2 and Explainable AI Using Grad-CAM"	Rashmi et al. [21	2023	Federated CNN with visual explainability	BrainCrossFed CNN
A Robust Distributed Deep Learning Approach to Detect Alzheimer's Disease from MRI Images"	Ghosh et al. [17]	2024	Distributed CNN	Multi-hospital AD detection

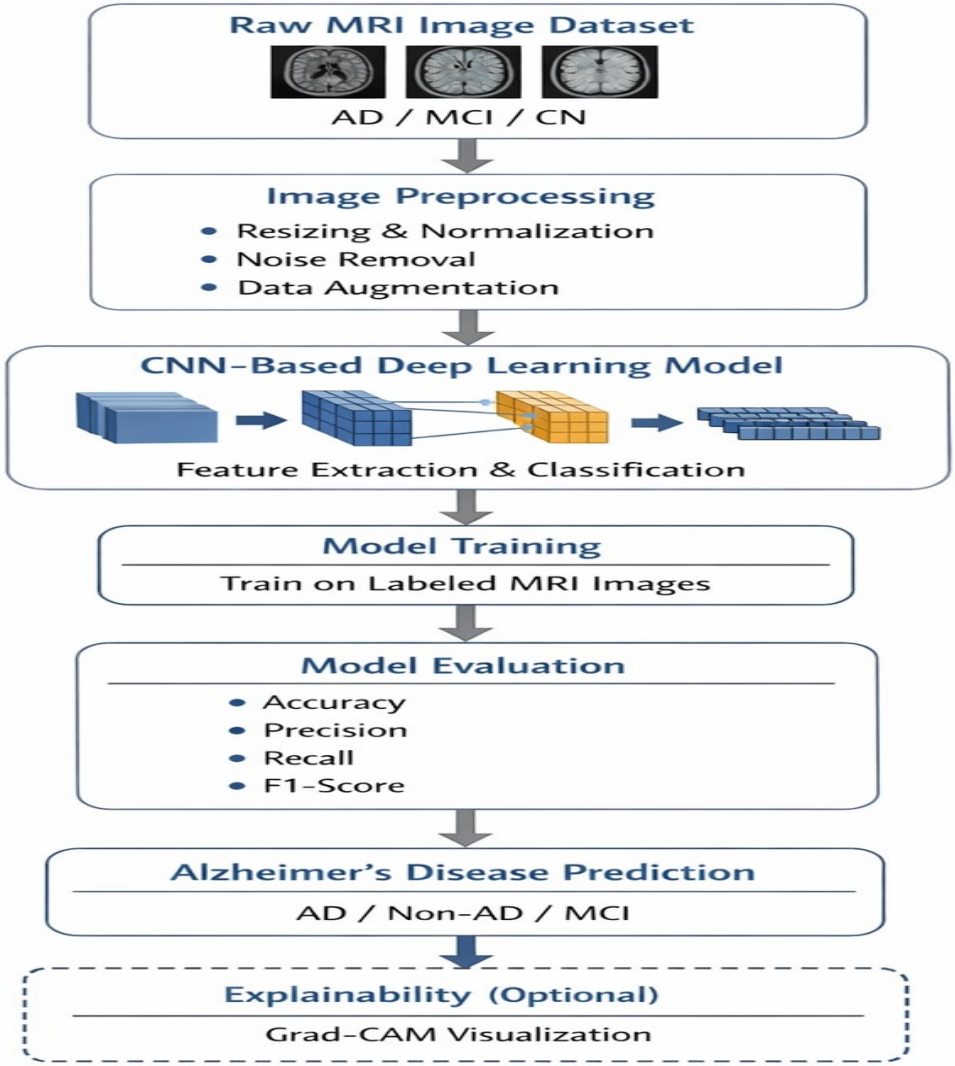
Data Set

- OASIS-3 dataset used for Alzheimer's Disease prediction
- Includes clinical, psychological, and MRI data, MRI images
- https://sites.wustl.edu/oasisbrains/home/oasis-3/
- MRI data contains brain structural information
- Multimodal data supports improved prediction accuracy
- Suitable for training and evaluation of AI models

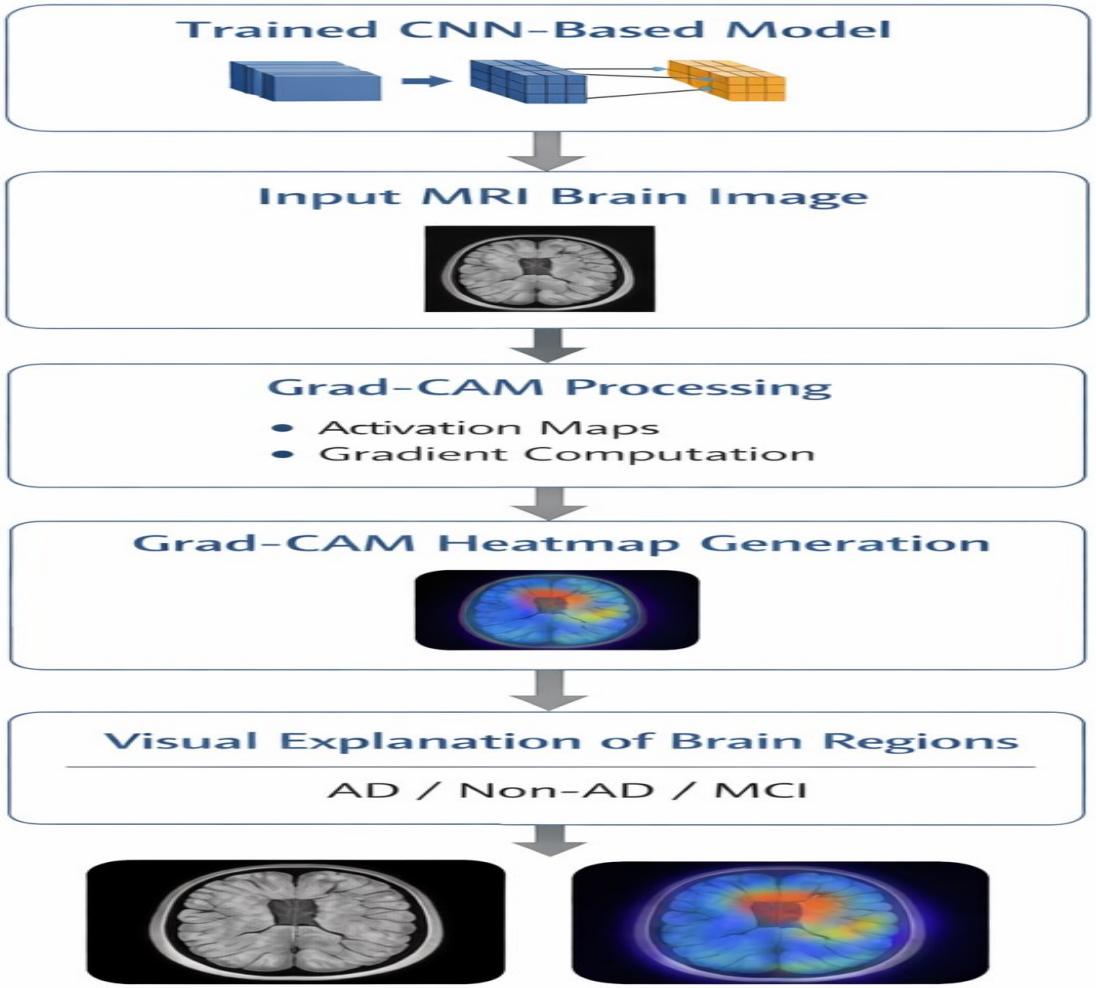
Proposed Method

- Use of raw MRI image datasets for Alzheimer's Disease prediction
- Implementation of CNN-based deep learning for spatial and structural feature learning
- Training of models within a Federated Learning framework to preserve data privacy
- Application of Grad-CAM for visual and interpretable predictions
- Development of a scalable and clinically reliable diagnostic system

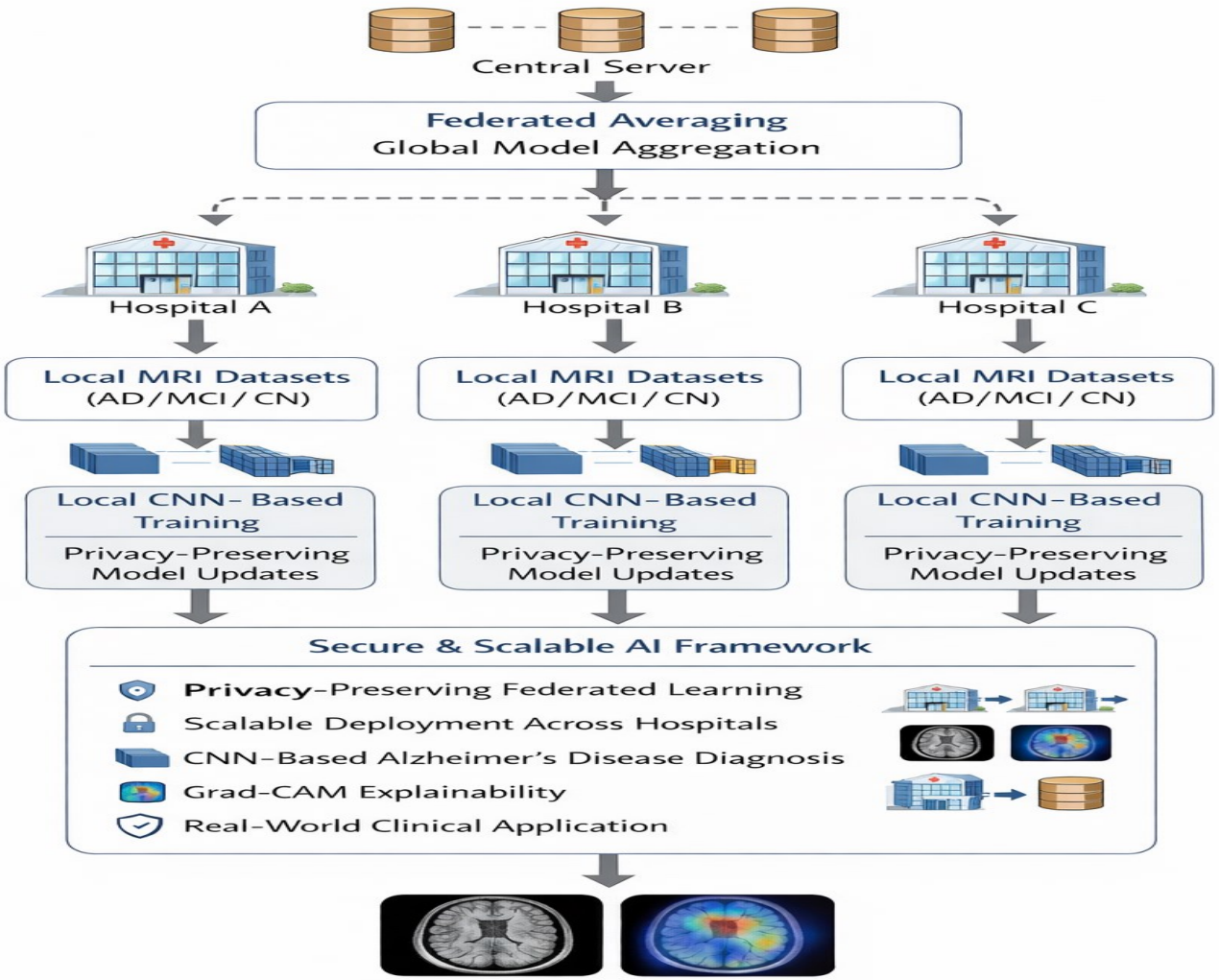
To improve Alzheimer's Disease prediction accuracy using raw MRI images and CNN-based deep learning


Raw MRI Image Dataset
AD / MCI / CN
Image Preprocessing
Resizing & Normalization
Noise Removal
Data Augmentation
CNN-Based Deep Learning Model
Feature Extraction & Classification
Model Training
Train on Labeled MRI Images
Model Evaluation
Accuracy
Precision
Recall
F1-Score
Alzheimer's Disease Prediction
AD / Non-AD / MCI
Explainability (Optional)
Grad-CAM Visualization

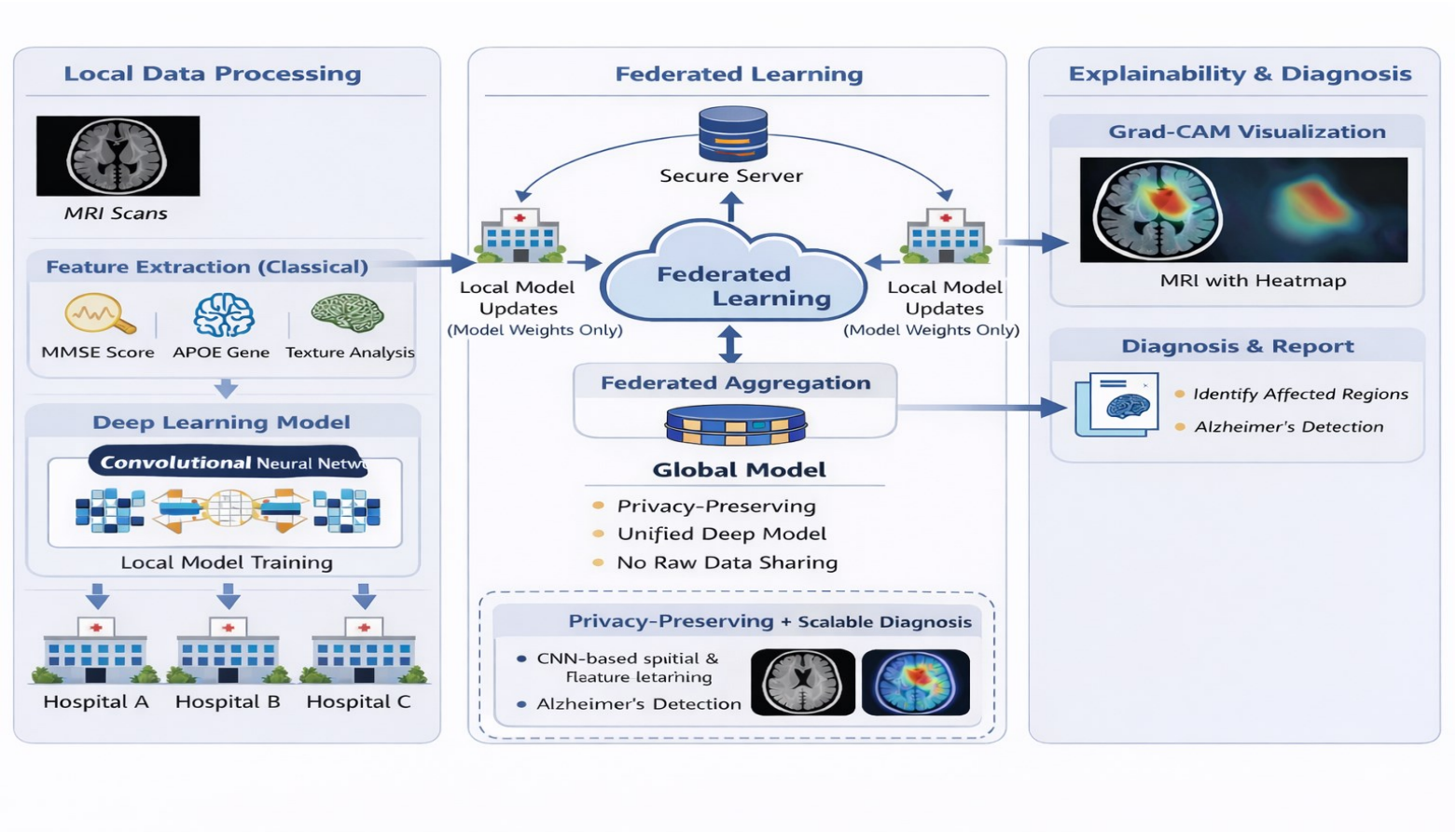
To enhance model interpretability through Grad-CAM visual explanations of brain regions


Trained CNN-Based Model
Input MRI Brain Image
Grad-CAM Processing
• Activation Maps
• Gradient Computation
Grad-CAM Heatmap Generation
Visual Explanation of Brain Regions
AD / Non-AD / MCI

To develop a privacy-preserving and scalable framework suitable for real-world clinical use


Central Server
Federated Averaging
Global Model Aggregation
Hospital A
Hospital B
Hospital C
Local MRI Datasets (AD/MCI/CN)
Local MRI Datasets (AD/MCI/CN)
Local MRI Datasets (AD/MCI/CN)
Local CNN-Based Training
Privacy-Preserving Model Updates
Local CNN-Based Training
Privacy-Preserving Model Updates
Local CNN-Based Training
Privacy-Preserving Model Updates
Secure & Scalable AI Framework
Privacy-Preserving Federated Learning
Scalable Deployment Across Hospitals
CNN-Based Alzheimer's Disease Diagnosis
Grad-CAM Explainability
Real-World Clinical Application

Architecture Diagram



Working

- MRI brain scans are collected and preprocessed locally at each hospital. Classical features and raw image data are handled without sharing patient data, ensuring data privacy.
- Each hospital trains a deep Convolutional Neural Network (CNN) using its own MRI dataset to learn spatial and structural brain features related to Alzheimer's Disease.
- Only trained model weights are sent to a secure central server. Raw MRI images never leave the hospital, maintaining strict patient confidentiality.
- The central server aggregates updates from all hospitals to form a robust global model that generalizes well across diverse clinical data sources.
- The final model provides Alzheimer's detection along with Grad-CAM heatmaps, highlighting affected brain regions to support transparent and trustworthy clinical decision-making.

Algorithm

- **Global Model Initialization:-**

Initialize a global Random Forest (RF) model at the central server with parameters V_0.

- **Client Selection:-**

In each communication round t, randomly select a subset of available client devices R_t

- **Global Model Distribution:-**

Broadcast the current global model V_{t-1} to all selected clients.

Algorithm

- **Local Model Training:-**

Each client trains the received RF model using its local multimodal dataset. Local model parameters are updated without sharing raw patient data.

- **Local Update Transmission:-**

Clients send only the updated model parameters to the central server.

- **Global Model Aggregation:-**

Aggregate received local updates using the **Federated Averaging (FedAvg)** algorithm. Generate an updated global model V_t.

- **Iteration**

Repeat Steps 2–6 for a predefined number of rounds or until convergence.

Algorithm

- **Explainable AI Integration**

Apply **SHAP Tree Explainer** on the trained global RF model. Compute feature importance values for each prediction.

- **Result Interpretation**

Visualize feature influence using SHAP plots. Provide transparent and interpretable Alzheimer's Disease predictions.

Results

Conclusion

- Phase-1 successfully demonstrates a privacy-preserving and explainable Alzheimer's prediction system
- The existing paper provides a strong and validated baseline
- Phase-2 extends the literature by introducing deep federated learning
- The proposed extension improves accuracy, interpretability, and clinical applicability
- The framework is suitable for future real-world clinical decision support systems

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3. (2022). *Alzheimer's: Death of Key Brain Cells Causes Daytime Sleepiness*. Accessed: Jul. 15, 2022. [Online]. Available: https://www.medicalnewstoday.com/articles/326073
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6. J. F. Raisa, S. Jahan, and M. S. Kaiser, ‘‘A cyber-physical fusion system for stress detection using multimodal and social media data,’’ in *Proc. 4th Int. Conf. Ind. Revolution Beyond*, S. Hossain, M. S. Hossain, M. S. Kaiser, S. P. Majumder, and K. Ray, Eds., Singapore: Springer, 2022, pp. 615–627.

7. N. Basnin, T. Mahmud, R. U. Islam, and K. Andersson, ‘‘An evolutionary federated learning approach to diagnose Alzheimer’s disease under uncertainty,’’ *Diagnostics*, vol. 15, no. 1, p. 80, Jan. 2025.